

Odprta koda – Domača naloga 1

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Delo s spremenljivkami

1.

```
val ime: String = "Ana"
```

2.

```
val letoRojstva: Int = 1990
```

```
val leto = 2025
```

```
var starost: Int = leto - letoRojstva
```

```
println("Starost: $starost")
```

3.

```
val starostDouble: Double = starost.toDouble()
```

```
println("Starost kot Double: $starostDouble")
```

4.

```
val nespremenljiva = 10
```

```
nespremenljiva = 20 // Napaka
```

5.

```
var spremenljiva = 5
```

```
spremenljiva = 10 // Dovoljeno
```

```
val konstanta = 3.14
```

```
konstanta = 2.71 // Napaka
```

6.

```
val cena: Float = 12.99f
```

```
val zaokrozenaCena = cena.toInt()
```

```
println("Zaokrožena cena: $zaokrozenaCena")
```

7.

```
val a = 10.0
```

```
val b = 20.0
```

```
val c = 30.0
```

```
val povprecje = (a + b + c) / 3
```

```
println("Povprečje: $povprecje")
```

8.

```
print("Vpiši ime: ")
```

```
val imev = readLine() ?: ""
```

```
println("Tvoje ime je $imev.")
```

9.

```
println("Dolžina imena: ${ime.length}")
```

10.

```
val prviZnak = if (ime.isNotEmpty()) ime[0] else ' '
```

```
println("Prvi znak imena: $prviZnak")
```

Operatorji

Aritmetični

1.

```
println(5 + 3)
```

2.

```
val stevilo = 7
```

```
if (stevilo % 2 == 0) println("Sodo") else println("Liho")
```

3.

```
val x = 10
```

```
val y = 20
```

```
val z = 30
```

```
val povp = (x + y + z).toDouble() / 3
```

```
println(povp)
```

4.

```
print("Dolžina: ")
```

```
val dolzina = readLine()?.toDoubleOrNull() ?: 0.0
```

```
print("Širina: ")
```

```
val sirina = readLine()?.toDoubleOrNull() ?: 0.0
```

```
println("Površina: ${dolzina * sirina}")
```

5.

```
val celsius = 25.0
```

```
val fahrenheit = celsius * 9 / 5 + 32
```

```
println("$celsius°C = $fahrenheit°F")
```

6.

```
val rezultat = 10.0 / 3.0
```

```
println("Rezultat: ${String.format("%.2f", rezultat)}")
```

7.

```
var in = 5
```

```
in++
```

```
println(in)
```

```
in--
```

```
println(in)
```

8.

```
val stev = 4
```

```
println(stev * stev)
```

9.

```
println(10 % 3)
```

10.

```
val doubleRez = 9.99
```

```
val intRez = doubleRez.toInt()
```

```
println(intRez)
```

Primerjalni

1.

```
val p = 15
```

```
val q = 20
```

```
if (p > q) println("p je večje") else println("q je večje")
```

2.

```
val x = 10
```

```
val y = 10
```

```
println(x == y)
```

3.

```
val starost = 20
```

```
if (starost >= 18) println("polnoleten")
```

4.

```
val niz1 = "Kotlin"
```

```
val niz2 = "Java"
```

```
println(niz1.length > niz2.length)
```

5.

```
val podatek = 5
```

```
println(podatek != 0)
```

6.

```
println(niz1 == niz2)
```

7.

```
val a = 1
```

```
val b = 2
```

```
val c = 3
```

```
println(a < b && b < c)
```

8.

```
val score = 95
```

```
if (score in 90..100) println("Odlično")
```

9.

```
println('a' < 'z')
```

10.

```
val datum1 = "2025-01-01"
```

```
val datum2 = "2024-12-31"
```

```
println(datum1 > datum2)
```

Logični

1.

```
val stevilo = 33
```

```
if (stevilo % 3 == 0 && stevilo % 11 == 0) println("Deljivo s 3 in 11")
```

2.

```
if (stevilo % 3 == 0 || stevilo % 11 == 0) println("Deljivo s 3 ali 11")
```

3.

```
val jeVključeno = false
```

```
if (!jeVključeno) println("Izključeno")
```

4.

```
val starost = 30
```

```
if (starost in 18..65) println("Delovno aktiven")
```

5.

```
val temp = 22
```

```
if ((temp > 0 && temp < 10) || (temp > 20 && temp < 30)) println("Zmerno")
```

6.

```
val znak = '5'
```

```
if (!znak.isLetter()) println("Ni črka")
```

7.

```
val nizA = "Test"
```

```
val nizB = ""
```

```
if (nizA.isEmpty() || nizB.isEmpty()) println("Vsaj en niz je prazen")
```

8.

```
fun pozitivnoInSodo(n: Int) = n > 0 && n % 2 == 0
```

```
println(pozitivnoInSodo(4))
```

9.

```
val imaVoznisko = true
```

```
val jePolnoleten = true
```

```
if (imaVoznisko && jePolnoleten) println("Lahko vozi")
```

10.

```
val a = 5
```

```
val b = 10
```

```
if (!(a > b)) println("a NI večje od b")
```

Pogojni stavki in zanke

1.

```
val starost = 16
```

```
if (starost >= 18) println("Polnoleten") else println("Ni polnoleten")
```

2.

```
val dan = 3
```

```
when (dan) {
```

```
    1 -> println("Ponedeljek")
```

```
    2 -> println("Torek")
```

```
    3 -> println("Sreda")
```

```
    4 -> println("Četrtek")
```

```
    5 -> println("Petek")
    else -> println("Vikend")
}
```

3.

```
val stev = -5

when {
    stev > 0 -> println("Pozitivna")
    stev < 0 -> println("Negativna")
    else -> println("Nič")
}
```

4.

```
for (i in 1..10) print("$i ")
println()
```

5.

```
var i = 1

while (i <= 5) {
    print("$i ")
    i++
}

println()
```

6.

```
var vsota = 0

for (j in 1..100) vsota += j

println("Vsota: $vsota")
```

7.

```
var count = 0

for (k in 1..50) if (k % 3 == 0) count++
```

```
println("Deljivih s 3: $count")
```

8.

```
for (l in 1..10) {  
    if (l == 5) break  
    print("$l ")  
}  
  
println()
```

9.

```
for (m in 1..10) {  
    if (m % 2 == 0) continue  
    print("$m ")  
}  
  
println()
```

10.

```
val imena = listOf("Ana", "Bojan", "Marko")  
  
for (ime in imena) println(ime)
```

Funkcije

1.

```
fun pozdravi(ime: String) {  
    println("Pozdravljen, $ime")  
}
```

2.

```
fun kvadrat(stevilo: Int): Int = stevilo * stevilo
```

3.

```
fun obseg(polmer: Double): Double = 2 * Math.PI * polmer
```

4.

```
fun jeSodo(stevilo: Int): Boolean = stevilo % 2 == 0
```


5.

```
fun vecji(a: Int, b: Int): Int = if (a > b) a else b
```

6.

```
fun najvecji(a: Int, b: Int, c: Int): Int = maxOf(a, b, c)
```

7.

```
fun nakljucnoStevilo(): Int = (1..100).random()
```

8.

```
fun zdruziNiza(a: String, b: String): String = a + b
```

9.

```
fun pozdrav(ime: String, ura: Int = 12) {
```

```
    when (ura) {
```

```
        in 0..11 -> println("Dobro jutro, $ime")
```

```
        in 12..17 -> println("Dober dan, $ime")
```

```
        else -> println("Dober večer, $ime")
```

```
    }
```

```
}
```

```
fun main() {
```

```
    pozdravi("Ana")
```

```
    println(kvadrat(4))
```

```
    println(obseg(21.0))
```

```
    println(jeSodo(7))
```

```
    println(vecji(10, 20))
```

```
    println(najvecji(3, 7, 5))
```

```
    println(nakljucnoStevilo())
```

```
    println(zdruziNiza("Kotlin ", "je super"))
```

```
    pozdrav("Ana", ura = 10)
```

```
}
```

Razredi in podatkovni razredi

1.

```
class Oseba(val ime: String, var starost: Int) {
```

2.

```
fun predstaviSe() {
```

```
    println("Živjo, ime mi je $ime in star sem $starost let.")
```

```
}
```

4.

```
override fun toString(): String {
```

```
    return "Oseba(ime='$ime', starost=$starost)"
```

```
}
```

```
}
```

3.

```
fun main() {
```

```
    val oseba1 = Oseba("Ana", 25)
```

```
    val oseba2 = Oseba("Bojan", 30)
```

```
    val oseba3 = Oseba("Marko", 22)
```

```
    oseba1.predstaviSe()
```

```
    oseba2.predstaviSe()
```

```
    oseba3.predstaviSe()
```

4. Testiraj toString()

```
    println(oseba1)
```

```
}
```

Polja

1.

```
val stevila = listOf(1, 2, 3, 4, 5)
```

2.

```
val mutableSeznam = mutableListOf(1, 2, 3)
```

```
mutableSeznam.add(4)
```

3.

```
mutableSeznam.remove(2)
```

4.

```
println(mutableSeznam.maxOrNull())
```

5.

```
mutableSeznam.forEach { println(it) }
```

6.

```
val sode = mutableSeznam.filter { it % 2 == 0 }
```

```
println(sode)
```

7.

```
val neurejen = listOf(5, 1, 9, 3)
```

```
val urejen = neurejen.sorted()
```

```
println(urejen)
```

8.

```
println(neurejen.contains(9))
```

9.

```
val besede = listOf("jabolko", "hruška", "banana")
```

```
val niz = besede.joinToString(", ")
```

```
println(niz)
```

Projekti

1.

```
1 import kotlin.random.Random
2
3 data class Artikel(val naziv: String, val vKosarici: Boolean = false) 4 Usages
4
5 class NakupovalniSeznam { 1 Usage
6     private val seznam = mutableListOf<Artikel>() 11 Usages
7
8     fun dodaj(naziv: String) { 1 Usage
9         seznam.add(Artikel(naziv))
10        println("Dodano: $naziv")
11    }
12
13    fun odstrani(naziv: String) { 1 Usage
14        val odstranjen = seznam.removeIf { it.naziv.equals(other = naziv, ignoreCase = true) }
15        if (odstranjen) println("Odstranjeno: $naziv")
16        else println("Artikel ne obstaja")
17    }
18
19    fun spremeni(stari: String, novi: String) { 1 Usage
20        val index = seznam.indexOfFirst { it.naziv.equals(other = stari, ignoreCase = true) }
21        if (index != -1) {
22            seznam[index] = Artikel(naziv = novi, seznam[index].vKosarici)
23            println("$spremenjeno: $stari -> $novi")
24        } else println("Artikel ne obstaja")
25    }
26
27    fun vKosarico(naziv: String) { 1 Usage
28        val index = seznam.indexOfFirst { it.naziv.equals(other = naziv, ignoreCase = true) }
29        if (index != -1) {
30            seznam[index] = Artikel(seznam[index].naziv, vKosarici = true)
31            println("$naziv je zdaj v kosarici")
32        } else println("Artikel ne obstaja")
33    }
34 }
```

```

35 fun izpisi() { 1 Usage
36     if (seznam.isEmpty()) println("Seznam je prazen")
37     else {
38         println("\nNakupovalni seznam")
39         for ((i, artikel) in seznam.withIndex()) {
40             val ikona = if (artikel.vKosarici) "[x]" else "[ ]"
41             println("${i+1}. $ikona ${artikel.naziv}")
42         }
43     }
44 }
45
46 fun vsebuje(naziv: String): Boolean { 1 Usage
47     return seznam.any { it.naziv.equals(other = naziv, ignoreCase = true) }
48 }
49
50
51 fun main() {
52     val ns = NakupovalniSeznam()
53     println("Ukazi: dodaj, odstrani, spremeni, kosarica, izpisi, vsebuje, konec")
54
55     while (true) {
56         print("> ")
57         when (readlnOrNull()?.trim()?.lowercase()) {
58             "dodaj" -> {
59                 print("Naziv: ")
60                 val n = readlnOrNull()?.trim() ?: continue
61                 ns.dodaj(naziv = n)
62             }
63             "odstrani" -> {
64                 print("Naziv: ")
65                 val n = readlnOrNull()?.trim() ?: continue
66                 ns.odstrani(naziv = n)
67             }
68             "spremeni" -> {
69                 print("Stari naziv: ")
70                 val s = readlnOrNull()?.trim() ?: continue
71                 print("Novi naziv: ")
72                 val n = readlnOrNull()?.trim() ?: continue
73                 ns.spremeni(stari = s, novi = n)
74             }
75             "kosarica" -> {
76                 print("Kateri artikel: ")
77                 val n = readlnOrNull()?.trim() ?: continue
78                 ns.vKosarico(naziv = n)
79             }
80             "izpisi" -> ns.izpisi()
81             "vsebuje" -> {
82                 print("Preveri: ")
83                 val n = readlnOrNull()?.trim() ?: continue
84                 println(if (ns.vsebuje(naziv = n)) "Artikel obstaja" else "Ne obstaja")
85             }
86             "konec" -> break
87             else -> println("Neveljaven ukaz")
88         }
89     }
90 }

```

2.

```
1 import kotlin.random.Random
2
3 data class PasswordRule(val length: Int, val useNumbers: Boolean, val useSymbols: Boolean) 1 Usage
4
5 fun main() {
6     print("Vnesi dolžino gesla: ")
7     val dolzina = readlnOrNull()?.toIntOrNull() ?: return
8     if (dolzina < 4) {
9         println("Geslo naj ima vsaj 4 znake")
10        return
11    }
12
13    print("Vključi številke? (d/n): ")
14    val stevilke = readlnOrNull()?.lowercase() == "d"
15    print("Vključi posebne znake? (d/n): ")
16    val simboli = readlnOrNull()?.lowercase() == "d"
17
18    val pravilo = PasswordRule(length = dolzina, useNumbers = stevilke, useSymbols = simboli)
19
20    var nabor = "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ"
21    if (pravilo.useNumbers) nabor += "0123456789"
22    if (pravilo.useSymbols) nabor += "!@#\$%&*?-_+~"
23
24    val geslo = StringBuilder()
25    repeat(times = pravilo.length) {
26        val randomIndex = Random.nextInt(until = nabor.length)
27        geslo.append(nabor[randomIndex])
28    }
29
30    println("Generirano geslo: $geslo")
31 }
```

3.

```
1 import kotlin.random.Random
2
3 data class Question(val text: String, val answer: String) 3 Usages
4
5 fun main() {
6     val questions = listOf(
7         Question(text = "Glavno mesto Slovenije?", answer = "Ljubljana"),
8         Question(text = "Koliko je 5 + 7?", answer = "12"),
9         Question(text = "Kateri jezik se uporablja za Android?", answer = "Kotlin")
10    )
11    println("MINI KVIZ")
12    var tocke = 0
13    var stVprasanj = 0
14
15    while (true) {
16        val q = questions.random()
17        println("Vprašanje: ${q.text}")
18        print("Tvoj odgovor: ")
19        val userAnswer = readlnOrNull()?.trim() ?: ""
20
21        if (userAnswer.equals(other = "konec", ignoreCase = true)) {
22            println("Konec kviza")
23            break
24        }
25
26        if (userAnswer.lowercase() == q.answer.lowercase()) {
27            println("Pravilno!")
28            tocke++
29        } else {
30            println("Napačno. Pravilen odgovor je: ${q.answer}")
31        }
32        stVprasanj++
33        println()
34    }
35    println("Pravilnih: $tocke od $stVprasanj")
36    if (stVprasanj > 0) {
37        println("Uspešnost: ${tocke * 100 / stVprasanj}%")
38    }
39 }
```